

LECTURE 02

INEQUALITIES

WORKING RULES FOR INEQUALITY

Addition:

$$\begin{array}{l} 2 < 4 \\ 2 + 2 < 4 + 2 \\ 4 < 6 \end{array} \quad \text{If } a < b \text{ then } a + c < b + c$$

Subtraction:

$$\begin{array}{l} 2 < 4 \\ 2 - 2 < 4 - 2 \\ 0 < 2 \end{array} \quad \text{If } a < b \text{ then } a - c < b - c$$

CONT...

Multiplication:

Multiplying by “2”

$$\begin{aligned}2 &< 4 \\2 \times 2 &< 4 \times 2 \\4 &< 8\end{aligned}$$

Inequality will remain same
when we multiply with
positive number

Multiplying by “-2”

$$\begin{aligned}2 &< 4 \\2 \times (-2) &< 4 \times (-2) \\-4 &> -8\end{aligned}$$

Inequality will change when
we multiply with negative
number

CONT...

Division:

Dividing by “2”

$$2 < 4$$

$$\frac{2}{2} < \frac{4}{2}$$

$$1 < 2$$

Inequality remains
constant

Dividing by “-2”

$$2 < 4$$

$$\frac{2}{-2} < \frac{4}{-2}$$

$$-1 > -2$$

Inequality changes if we
divide with negative
number

CONT...

Multiplicative Inverse:

Less than changes into greater than when we take multiplicative inverse.

$$2 < 4$$

$$\frac{1}{2} > \frac{1}{4}$$

EXAMPLE # 01 (LINEAR INEQUALITY)

Solve the inequality $4x + 3 > 2x - 5$.

Solution:

$$4x + 3 > 2x - 5$$

$$4x > 2x - 8$$

$$2x > -8$$

$$x > -4$$

Hence, the solution consist of all real numbers greater than -4, that is, the numbers in the infinite interval $(-4, \infty)$.

EXAMPLE # 02 (COMPOUND INEQUALITY)

Solve the inequality $-5 < \frac{4-3x}{2} < 1$.

Solution:

$$-5 < \frac{4-3x}{2} < 1$$

$$-10 < 4-3x < 2$$

$$-14 < -3x < -2$$

$$\frac{14}{3} > x > \frac{2}{3}$$

$$\frac{2}{3} < x < \frac{14}{3}$$

Hence, the solutions are the numbers in the open interval $(\frac{2}{3}, \frac{14}{3})$.

EXAMPLE # 03 (QUADRATIC INEQUALITY)

Solve $x^2 - 7x + 10 > 0$

Solution:

Since the inequality may be written as after factorization

$$(x - 5)(x - 2) > 0$$

It follows that x is a solution if and only if both factors $x - 5$ and $x - 2$ are positive, or both are negative.



CONT...

Evidently, both factors are positive if x is in the interval $(5, \infty)$ and both are negative if x is in $(-\infty, 2)$. Hence the solutions consist of all real numbers in the union $(-\infty, 2) \cup (5, \infty)$.

ABSOLUTE VALUE

The absolute value of a is defined as:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a \leq 0 \end{cases}$$

Properties:

➤ $|a| = |-a|$, $|ab| = |a||b|$, $-|a| \leq a \leq |a|$

➤ It can be shown that if b is any positive real number then

i. $|a| < b$ iff $-b < a < b$

ii. $|a| > b$ iff $a > b$ or $a < -b$

iii. $|a| = b$ iff $a = b$ or $a = -b$

The triangle inequality

$$|a + b| \leq |a| + |b|$$

EXAMPLE # 04

Solve the inequality $|x - 3| < 0.1$

Solution:

$$\begin{aligned} -0.1 &< x - 3 < 0.1 \\ -0.1 + 3 &< x - 3 + 3 < 0.1 + 3 \\ 2.9 &< x < 3.1 \end{aligned}$$

Thus, the solutions are the real numbers in the interval $(2.9, 3.1)$.

EXAMPLE # 05

Solve the inequality $|2x - 7| > 3$

Solution:

x is a solution of $|2x - 7| > 3$ if and only if either

$$2x - 7 > 3 \quad \text{or} \quad 2x - 7 < -3$$

$$2x > 10 \quad \text{or} \quad 2x < 4$$

$$x > 5 \quad \text{or} \quad x < 2$$

Hence the solutions of $|2x - 7| > 3$ are the numbers in the union $(-\infty, 2) \cup (5, \infty)$.

Solve the inequalities in Exercises 7–34 and express the solutions in terms of intervals.

7 $5x - 6 > 11$

8 $3x - 5 < 10$

9 $2 - 7x \leq 16$

10 $7 - 2x \geq -3$

11 $|2x + 1| > 5$

12 $|x + 2| < 1$

13 $3x + 2 < 5x - 8$

14 $2 + 7x < 3x - 10$

15 $12 \geq 5x - 3 > -7$

16 $5 > 2 - 9x > -4$

17 $-1 < \frac{3 - 7x}{4} \leq 6$

18 $0 \leq 4x - 1 \leq 2$

19 $\frac{5}{7 - 2x} > 0$

20 $\frac{4}{x^2 + 9} > 0$

21 $|x - 10| < 0.3$

22 $\left| \frac{2x + 3}{5} \right| < 2$

$$23 \quad \left| \frac{7 - 3x}{2} \right| \leq 1$$

$$24 \quad |3 - 11x| \geq 41$$

$$25 \quad |25x - 8| > 7$$

$$26 \quad |2x + 1| < 0$$

$$27 \quad 3x^2 + 5x - 2 < 0$$

$$28 \quad 2x^2 - 9x + 7 < 0$$

$$29 \quad 2x^2 + 9x + 4 \geq 0$$

$$30 \quad x^2 - 10x \leq 200$$

$$31 \quad \frac{1}{x^2} < 100$$

$$32 \quad 5 + \sqrt{x} < 1$$

$$33 \quad \frac{3x + 2}{2x - 7} \leq 0$$

$$34 \quad \frac{3}{x - 9} > \frac{2}{x + 2}$$

35 The relationship between the Fahrenheit and Celsius temperature scales is given by $C = (\frac{5}{9})(F - 32)$. If $60 \leq F \leq 80$, express the corresponding range for C in terms of an inequality.

- 36 In the study of electricity, Ohm's Law states that if R denotes the resistance of an object (in ohms), E the potential difference across the object (in volts), and I the current that flows through it (in amperes), then $R = E/I$ (see figure). If the voltage is 110, what values of the resistance will result in a current that does not exceed 10 amperes?

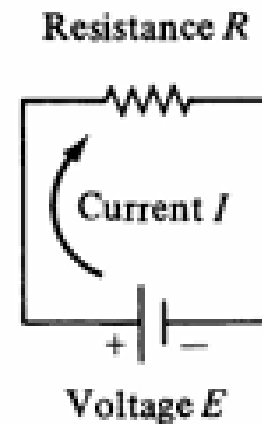


FIGURE FOR EXERCISE 36

- 37 According to Hooke's Law, the force F (in pounds) required to stretch a certain spring x inches beyond its natural length is given by $F = (4.5)x$ (see figure). If $10 \leq F \leq 18$, what is the corresponding range for x ?

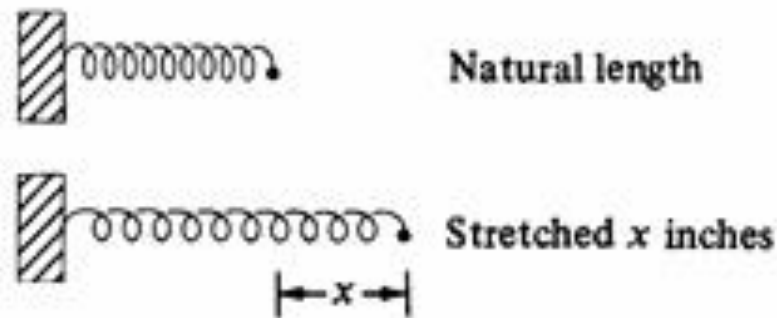


FIGURE FOR EXERCISE 37

- 38 Boyle's Law for a certain gas states that $pv = 200$, where p denotes the pressure (lb/in.²) and v denotes the volume (in.³). If $25 \leq v \leq 50$, what is the corresponding range for p ?
- 39 If a baseball is thrown straight upward from level ground with an initial velocity of 72 ft/sec, its altitude s (in feet) after t seconds is given by $s = -16t^2 + 72t$. For what values of t will the ball be at least 32 feet above the ground?
- 40 The period T (sec) of a simple pendulum of length l (cm) is given by $T = 2\pi\sqrt{l/g}$, where g is a physical constant. If, for the pendulum in a grandfather clock, $g = 980$ and $98 \leq l \leq 100$, what is the corresponding range for T ?